Repeatable station terms redistribute long-period response spectra across Japanese strong-motion sites

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Abstract

Regional hazard products average ground motion, yet repeatable station departures can alter local spectra. Applying them requires tests across event samples, ground-motion models and paths. Here we present an analysis of 222,664 K-NET and KiK-net surface records at eight MF2013-compatible RotD100 periods from 0.1 to 5.0 s. At 3.0 s, station terms from disjoint event groups correlate at 0.890; MF2013 and Zhao 2006 fields correlate at 0.770. Site and location variables reduce spatial-block root mean squared error by 12.6% across 1,628 stations. Source-class fields match the full field, whereas directional-sector correlations of 0.280–0.886 indicate path conditioning. Cross-validated multipliers span 0.624–1.338 and shift the median matched-station surface spectrum from 0.081 to 0.075 g. The repeatable average station field changes published spectra; source-specific corrections require path-conditioned validation.

Keywords: strong-motion records; station terms; path effects; response spectra; Japan

1 Introduction

Regional ground-motion models assign a common median to recordings with similar source, path and site descriptors. Repeated recordings at the same station often remain systematically above or below that median, motivating nonergodic source, site and path terms[1–5]. Fully nonergodic models for Japan already resolve spatially varying source, site and path adjustments, and three-dimensional simulations provide an additional constraint on basin-related site terms[6, 7].

Japan offers an unusually dense test of station-specific residuals. The J-SHIS/NIED Strong Ground Motion Flat File links K-NET and KiK-net records to earthquake sources, distances and site metadata[8–12]. The Morikawa–Fujiwara 2013 model, hereafter MF2013, includes a magnitude-distance backbone and site terms based on AVS30 and D1400[13, 14]. AVS30 is the average shear-wave velocity in the upper 30 m, and D1400 is the depth to the 1.4 km s^{-1} shear-wave velocity horizon. Their period-dependent coefficients provide a direct test of shallow-soil and deep-sedimentary contributions.

Long-period motion is sensitive to sedimentary depth and basin geometry. Observations and simulations in Japan show persistent amplification and surface-wave effects at periods relevant to large basins and long-period structures[15–18]. A station residual in this band can combine local response, unresolved path structure and sensor effects. Japan's provisional probabilistic response-spectrum maps provide an engineering-bedrock coordinate on which such a correction can be tested[19].

We evaluate station-field stability under changes in earthquake sample, ground-motion model and path distribution, then apply spatial-block out-of-fold predictions to matched J-SHIS response spectra. Event holdout and a Zhao 2006 calculation test sample and model dependence. Network, regional, event-influence and path-stratified analyses define the spatial and source-path range of the inferred field. Random event groups reproduce the average field, and site and location variables predict part of it at held-out stations. Directional and short-distance subsets show that source-specific corrections require path-conditioned validation. The spectrum calculation quantifies an average-path station adjustment on the published hazard coordinate.

2 Results

2.1 Surface-record residuals

The flatfile contains 333,808 records, including paired surface and borehole KiK-net sensors. The primary calculation retains only installations coded as ground surface. The MF2013 screen then yields 222,664 records from 1,880 surface stations and 1,737 earthquakes at each period. Among these stations, 1,628 have at least 20 records and enter the station-level analysis. Their distribution covers the Japanese islands, and the earthquake set includes crustal, interplate and intraplate sources (Fig. 1a).

The D1400 and AVS30 terms improve the residual fit most strongly between 1 and 3 s. Their reduction in mean absolute residual is 3.0% at 0.1 s and 14.0% at 1.0 s, reaches 20.0% and 21.4% at 2.0 and 3.0 s, and decreases to 16.2% at 5.0 s (Fig. 1b; Table 1; Supplementary Table 1). Every period is read directly from the flatfile RotD100 spectrum.

For each period, the residual is separated into a global intercept, an event term, a station term and a record remainder. The station component is larger at short periods and decreases towards 5.0 s, while the event and record components vary less across period (Fig. 1c). The global intercept is retained as a model offset and is excluded from the station correction.

2.2 Station terms recur across events

Station terms estimated from disjoint event groups remain strongly correlated. At 3.0 s, each test group contains about 347 earthquakes, and an average of 1,481 stations meet the train and test record thresholds. Separate two-way event-station decompositions of the training and test records give a mean train-test correlation of 0.890. Applying the training-event station term to the test-event station term reduces root mean squared error by 67.7% relative to zero. Correlations range from 0.871 to 0.917 across 0.1–5.0 s, and the period-specific reductions range from 65.6% to 73.5% (Fig. 1d; Supplementary Fig. 1; Supplementary Table 2).

The MF2013 site terms remove most of the long-period association with D1400. At 3.0 s, the event-adjusted station correlation with D1400 decreases from 0.751 for the magnitude-distance backbone to 0.194 after the D1400 and AVS30 terms are added (Fig. 2a,c). The corresponding correlation with basement depth decreases but remains positive (Fig. 2b). The residual field still contains basin or spatial structure beyond the two MF2013 site variables.

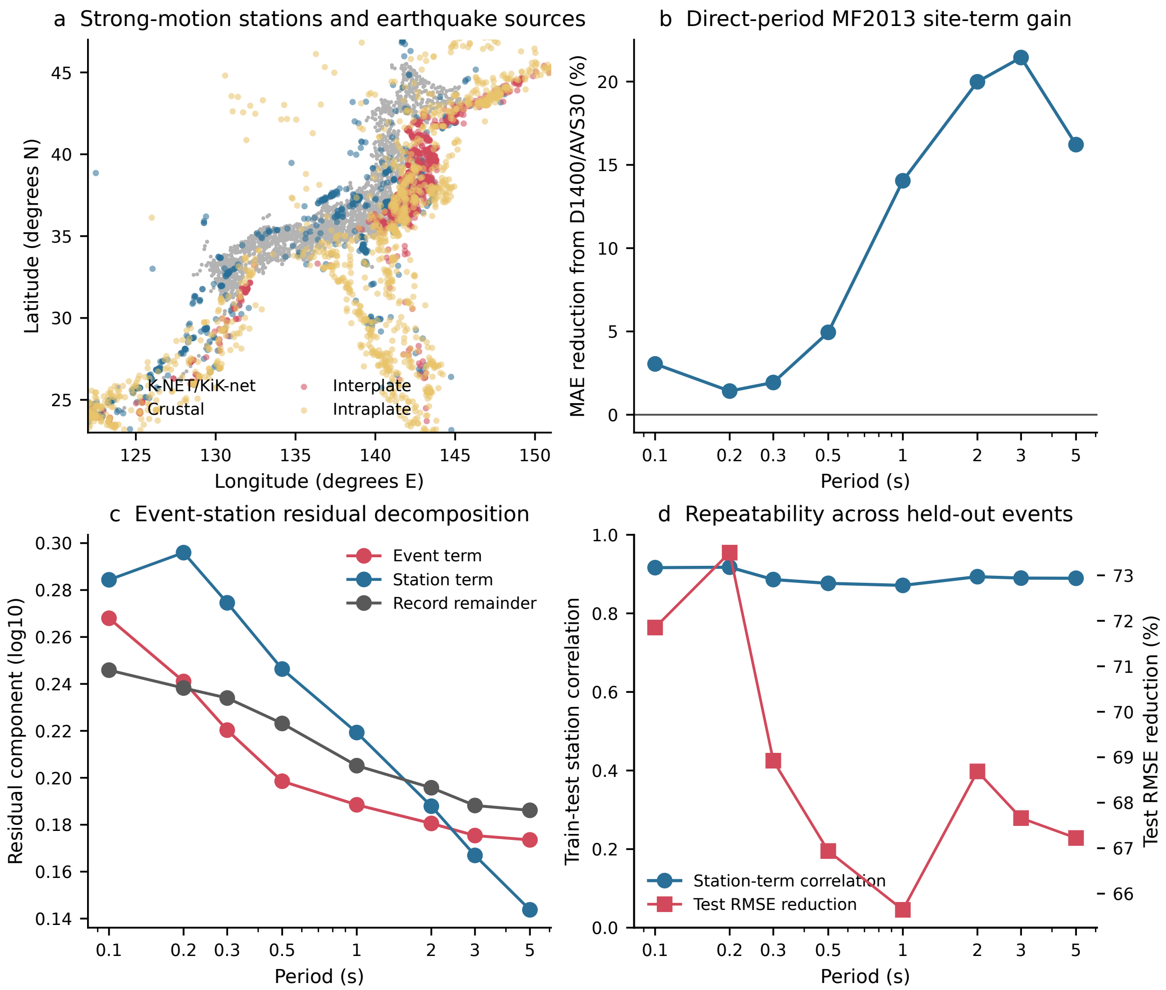

Figure 1. Data coverage and residual decomposition. **a**, K-NET and KiK-net stations and earthquake sources used in the residual calculation. **b**, Reduction in mean absolute residual after adding the MF2013 D1400 and AVS30 terms at directly observed periods. **c**, record-weighted standard deviations of the event and station terms and the root mean squared record remainder. **d**, repeatability of station terms across five held-out event groups. Blue circles show the mean train-test station-term correlation; red squares show the mean reduction in test root mean squared error relative to a zero station term.

An approximate application of the MF2013 anomalous-intensity term changes individual residuals but preserves the station field. At 3.0 s, primary and sensitivity station terms correlate at 0.971, and the D1400 correlation changes from 0.194 to 0.206 (Fig. 2d). The main period pattern is insensitive to this approximation.

2.3 Station patterns persist under a second model

We repeated the residual decomposition with the Zhao et al. 2006 model as implemented in OpenQuake[20, 21]. The source-class-specific active-crustal, interface and intraslab variants use the same surface observations, source classes, AVS30 values and documented shortest fault distances. At 3.0 s, the MF2013 and Zhao station terms correlate at 0.770 across 1,628 stations; the record-weighted correlation is 0.735, and the median absolute difference is 0.151 $\log_{10}$ units (Fig. 3; Supplementary Table 3). Their signs agree at 77.5% of stations. Correlations decrease from 0.953 at 0.1 s to 0.697 at 5.0 s, showing greater backbone dependence at long periods while

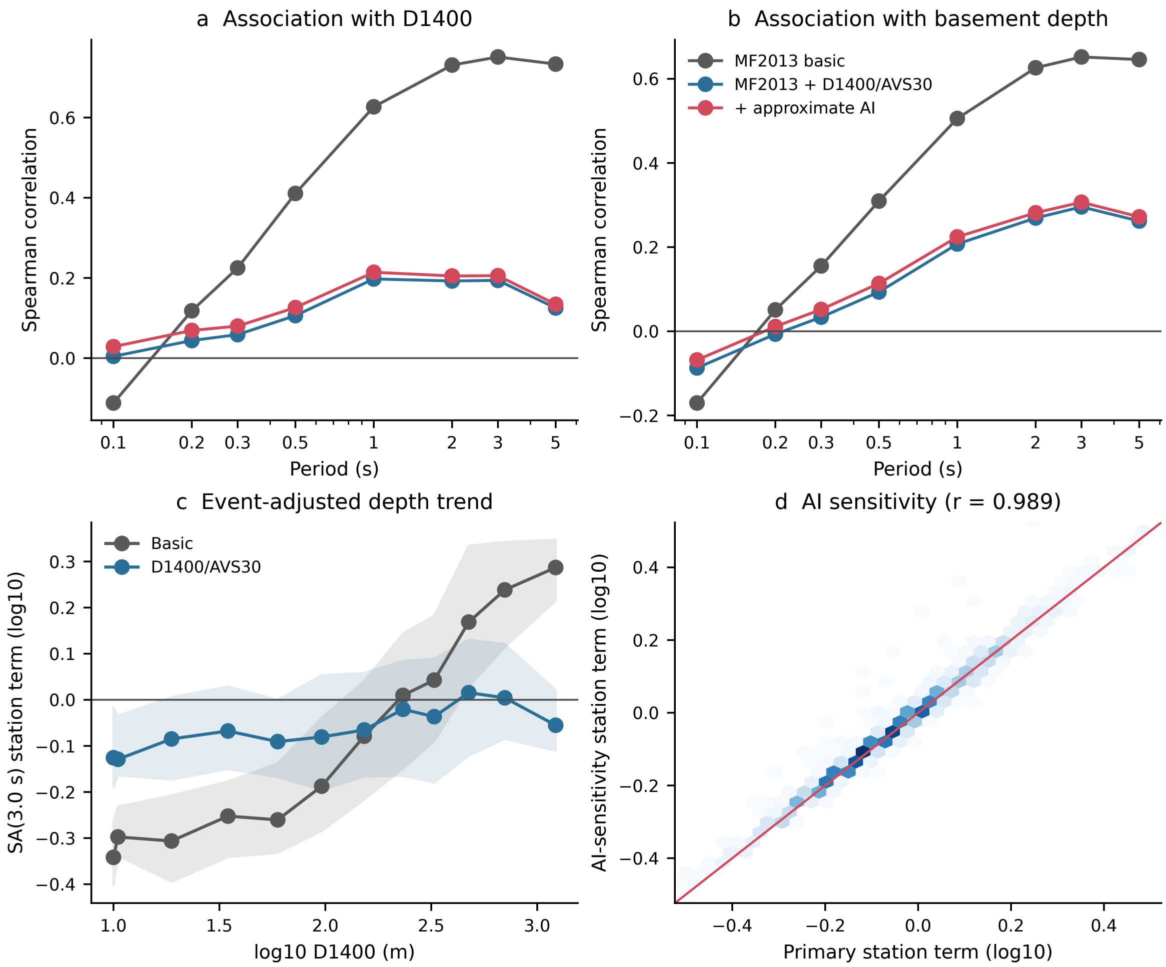

Figure 2. Event-adjusted station terms and sedimentary-depth variables. **a**, Spearman correlations between station terms and D1400. **b**, correlations with basement depth. Grey denotes the MF2013 magnitude-distance backbone, blue includes D1400 and AVS30, and red adds the rule-based anomalous-intensity sensitivity. **c**, binned SA(3.0 s) station terms against D1400 before and after the site terms; lines show bin medians and bands show interquartile ranges. **d**, comparison between primary and anomalous-intensity sensitivity station terms at 3.0 s.

retaining a common average ordering.

The Zhao residuals also repeat across held-out event groups. Their 3.0 s train-test correlation is 0.950, and the site-and-location model reduces spatial-block root mean squared error by 37.7%, with an observed-predicted correlation of 0.792. Both calculations use the same observations and validation design, so this comparison measures sensitivity to the ground-motion model.

2.4 Site and location variables predict held-out stations

The site-only model uses shallow velocities, sedimentary depths, elevation, sensor depth and network. At 3.0 s its spatial-block root mean squared error is 2.3% higher than the zero-term baseline. After longitude, latitude and volcanic-front distances are added, the site-and-location model reduces error by 12.6% and gives an observed-predicted correlation of 0.500. Error reductions across the five held-out blocks range from 10.2% to 34.1%. The site descriptors retain associations with station residuals; prediction beyond a spatial block also draws on location.

The site-and-location model improves on the zero baseline at every direct period. Root

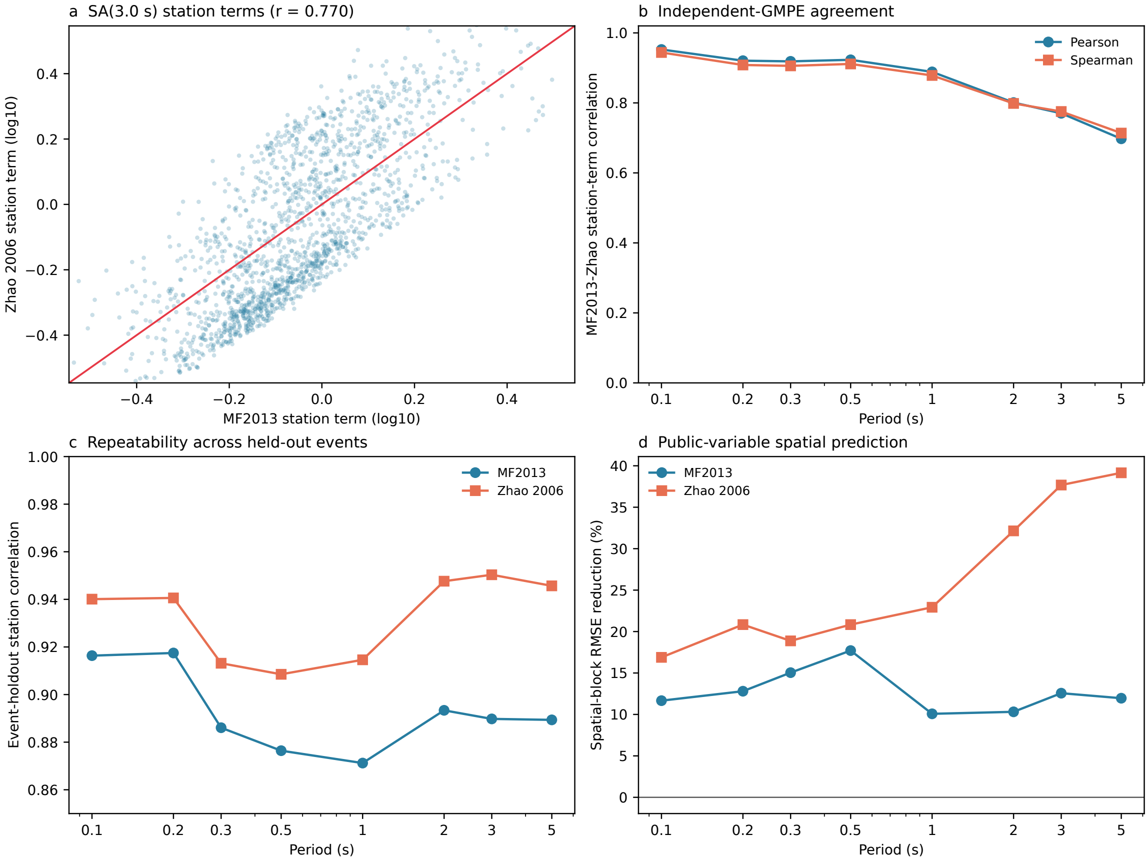

Figure 3. Ground-motion-model sensitivity of the station field. **a**, MF2013 and Zhao 2006 station terms at 3.0 s. **b**, station-term correlations across the eight direct periods. **c**, event-group repeatability for both residual models. **d**, spatial-block root mean squared error reduction for the site-and-location model. Both residual calculations use the same surface records.

Table 1. Direct period residual and prediction results

Period (s)	MAE gain (%)	D1400 correlation Basic	Site	Event holdout correlation	Spatial model RMSE gain (%)
0.1	3.0	-0.112	0.004	0.916	11.7
0.2	1.4	0.118	0.044	0.917	12.8
0.3	1.9	0.225	0.059	0.886	15.0
0.5	4.9	0.411	0.106	0.876	17.7
1.0	14.0	0.627	0.198	0.871	10.1
2.0	20.0	0.731	0.192	0.893	10.3
3.0	21.4	0.751	0.194	0.890	12.6
5.0	16.2	0.733	0.125	0.889	12.0

mean squared error reductions are 10.1%, 10.3%, 12.6% and 12.0% at 1.0, 2.0, 3.0 and 5.0 s, respectively (Fig. 4b; Supplementary Fig. 2; Supplementary Table 4). A common-feature model that excludes VS20, VS30 and network identity also transfers between K-NET and KiK-net. At 3.0 s it reduces error by 8.7% when K-NET is held out and by 20.5% when KiK-net is held out. Fixed macroregion tests are less uniform: the full model improves all five regions by 5.4% to 28.9%, whereas the common-feature sensitivity changes from −5.2% to 27.5% (Supplementary Fig. 3; Supplementary Table 5). Joint deletion of the ten highest-count events leaves station-term correlations above 0.9989 at every period and changes the 3.0 s field by 0.012 $\log_{10}$ units at its record-weighted 95th percentile (Supplementary Table 6).

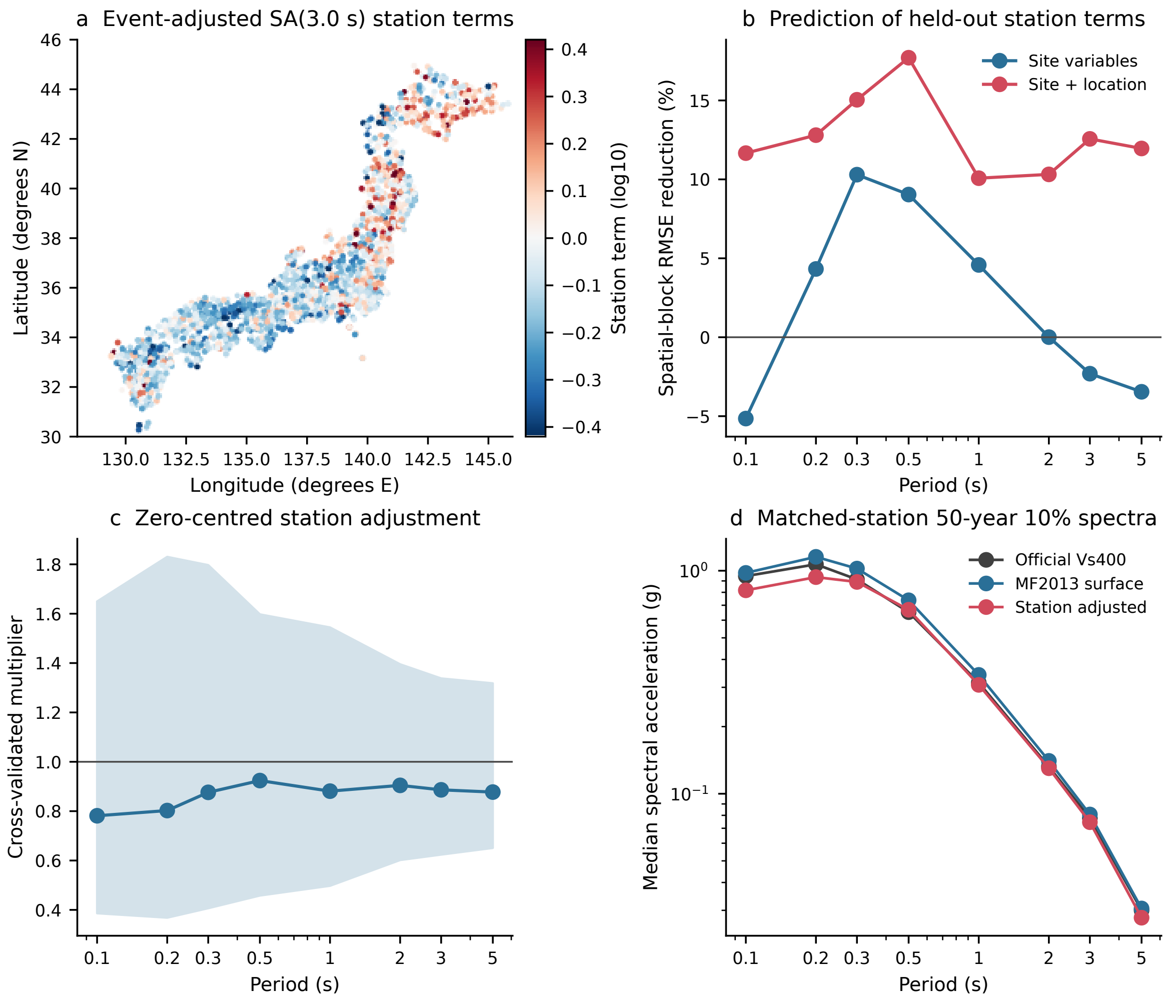

Figure 4. Prediction and response-spectrum propagation. **a**, event-adjusted SA(3.0 s) station terms at stations with at least 20 records. **b**, spatial-block root mean squared error reduction for models using site variables and site plus location variables. **c**, median and 5th–95th percentile range of zero-centred cross-validated station multipliers. **d**, matched-station median 50-year 10% response spectra on official $V_S = 400$ m s^{-1} engineering bedrock, after projection to the station surface condition, and after the station adjustment. All periods are calculated directly.

2.5 Path strata expose conditional transfer

Random event holdout preserves the broad path distribution seen by each station. We re-estimated station terms within four hypocentral-bearing sectors, four shortest-fault-distance bins and the three source classes. At 3.0 s, source-class fields correlate at 0.924–0.942 with the full field, and the 100–300 km distance fields correlate at 0.963–0.974. Directional-sector correlations span 0.280–0.886, while correlations within disjoint event halves of the same sectors span 0.762–0.967 (Fig. 5; Supplementary Table 7). High within-sector repeatability alongside low full-field agreement shows a systematic directional component.

Equal weighting across available path strata retains a weaker persistent component. At 1–3 s, models fitted to azimuth-, distance- and source-class-balanced fields reduce overall spatial-block error by 2.5%–9.6%. Several individual spatial blocks have negative gains, including the azimuth-balanced 3.0 s field (Supplementary Fig. 4; Supplementary Table 8). Public station variables predict an average component of the residual field. Prediction strength varies with path coverage.

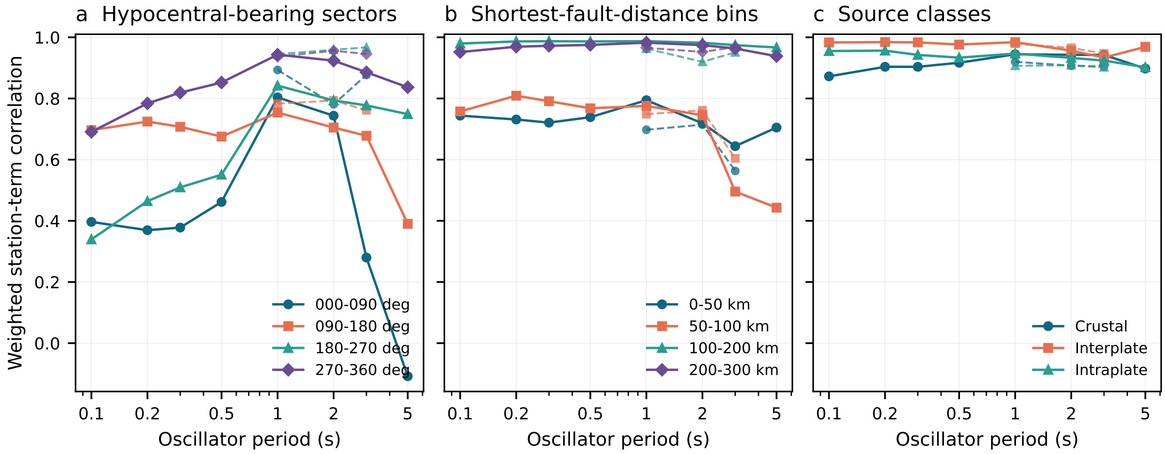

Figure 5. Path and source stratification of the station field. **a**, hypocentral-bearing sectors. **b**, shortest-fault-distance bins. **c**, source classes. Solid lines compare each stratum field with the full station field at all eight periods. Dashed lines show mean correlations between five pairs of disjoint event halves at 1.0, 2.0 and 3.0 s. Disconnected station-event graphs are solved and aligned by component before comparison.

2.6 Local spectra change in both directions

The predicted station terms are centred to zero in log space before propagation. At 3.0 s, their multipliers have 5th, 50th and 95th percentiles of 0.624, 0.886 and 1.338 (Fig. 4c). The corresponding local changes are −37.6%, −11.4% and +33.8%. At 5.0 s, the 5th and 95th percentiles are 0.651 and 1.318.

Official J-SHIS response-spectrum ordinates are defined on engineering bedrock with $V_S = 400$ m s^{-1}. At each matched station, we replace the MF2013 shallow-soil factor for 400 m s^{-1} with the factor for the station AVS30. For the 50-year 10% probability level at 3.0 s, the median official ordinate is 0.078 g and the median surface-reference ordinate is 0.081 g. Applying the cross-validated station multiplier changes the median to 0.075 g (Fig. 4d; Supplementary Fig. 5; Supplementary Table 9). The global MF2013 residual offset is excluded from this station correction.

Spatial-block errors provide a separate empirical prediction interval for each held block. At 3.0 s, the nominal 90% interval covers 89.0% of stations and 88.8% after record-count weighting; its median upper-to-lower multiplier span is 3.00 (Supplementary Fig. 6; Supplementary Table 10). These intervals quantify station-model error under the spatial validation design. Source recurrence, ground-motion-model uncertainty and the J-SHIS logic tree remain those of the published hazard product.

City-nearest matched stations illustrate the bidirectional effect. At 3.0 s, the cross-validated changes are −2.5% near Tokyo, −11.5% near Osaka, +31.2% near Sapporo, −11.0% near Sendai and −25.2% near Fukuoka (Fig. 6; Supplementary Table 11). Each value is a station-level example for the nearest analysed station.

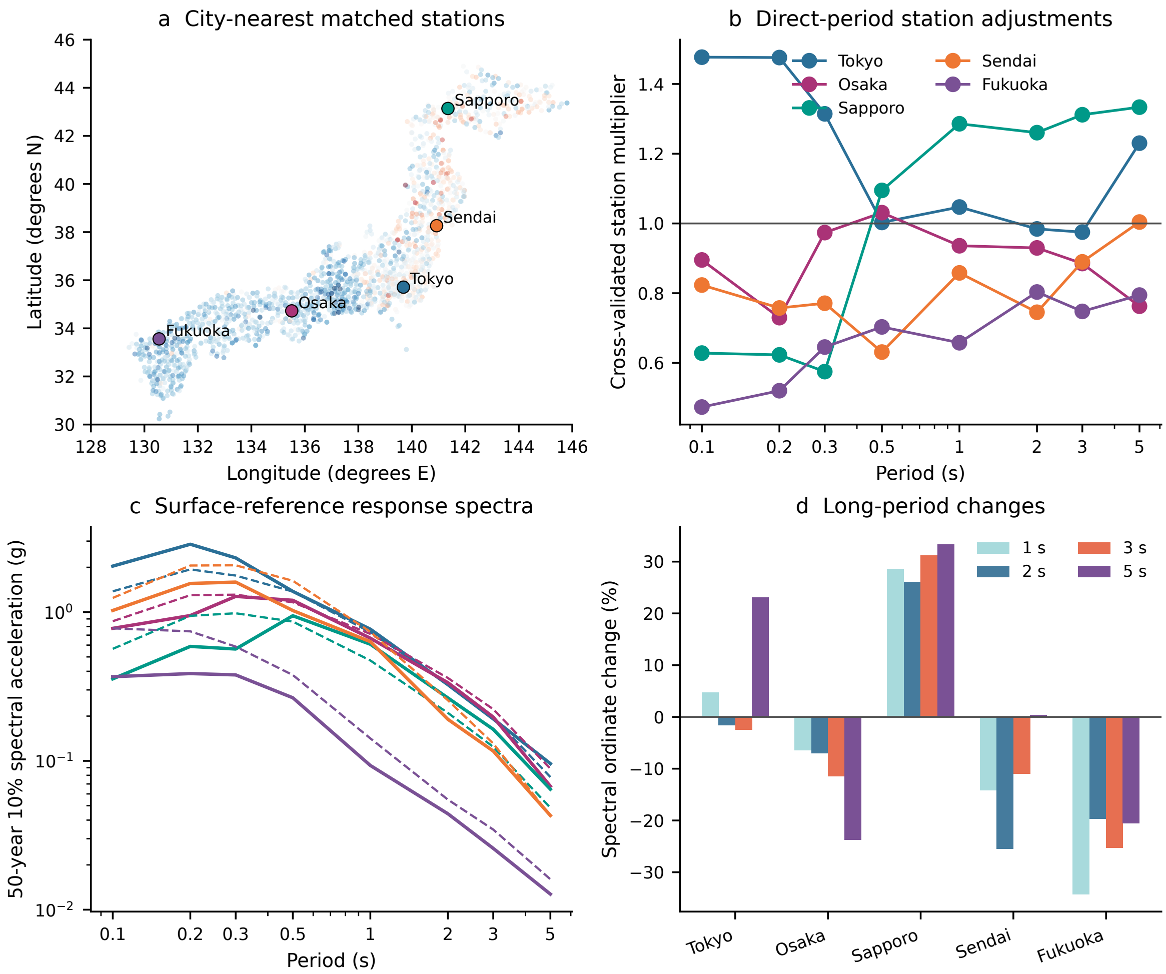

Figure 6. Illustrative city-nearest station cases. **a**, analysed stations nearest to five Japanese cities. **b**, direct-period cross-validated multipliers at the selected stations. **c**, 50-year 10% surface-reference response spectra before and after the station adjustment; dashed and solid lines denote the two spectra. **d**, percentage changes at 1.0, 2.0, 3.0 and 5.0 s. Distances to the selected stations are 2.4 km for Tokyo, 3.6 km for Osaka, 8.6 km for Sapporo, 5.0 km for Sendai and 14.1 km for Fukuoka.

3 Discussion

Fully nonergodic Japanese ground-motion models separate spatially varying source, site and path terms, and three-dimensional simulations constrain basin response[6, 7]. We evaluate a simpler station field derived from public flatfile variables and connect it to an official response-spectrum product. MF2013 D1400 and AVS30 terms remove much of the 1–5 s sedimentary-depth trend. The remaining field recurs across random event groups and retains a common ordering under Zhao 2006. Regional site-response models and residual-based basin analyses describe related spatial structure at broader scales[22, 23].

The response-component comparison separates amplitude coordinates from spatial structure. RotD100 matches the MF2013 horizontal-vector target. At 3.0 s, RotD50 and RotD100 station fields correlate at 0.9986 and differ by 0.018 $\log_{10}$ units at the 95th percentile. Their record-level median ratio is 0.819, and their remainder dispersions differ. Component choice has little effect on station ordering, whereas absolute residual levels and variability depend on the model-compatible coordinate.

Random event holdout measures repeatability under each station's broad observed path mixture. Path-stratified analyses then test how the field changes with direction, distance and source class. The 3.0 s field for the 000–090° sector is internally repeatable across disjoint event halves at 0.877 and correlates with the full field at 0.280. Source-class fields are more stable, and the 100–300 km distance fields closely follow the full ordering. The contrast identifies station-specific path structure, which fully nonergodic models represent through explicit path terms[6]. The station quantity used here is an average over the observed source-path distribution.

Public site and location variables recover part of that average field. The 3.0 s spatial-block gain is 12.6%, all five primary folds improve, and the common-feature model transfers in both directions between K-NET and KiK-net. Equal-stratum fields remain predictable in aggregate, with 1–3 s gains of 2.5%–9.6%. Several individual spatial folds deteriorate. The model supports interpolation within well-sampled parts of the existing network. New regions and source-path distributions require local records or an explicit path model.

The spectrum calculation keeps the official $V_S = 400$ m s^{-1} ordinate, the MF2013 surface projection and the zero-centred station adjustment as separate quantities. At 3.0 s, the central 90% of point multipliers spans 0.624–1.338; the empirical station-model interval has a median upper-to-lower factor of 3.00. The matched-mesh calculation retains source recurrence, aleatory variability and logic-tree aggregation from the published J-SHIS product[24–30]. The resulting values quantify average-path sensitivity. Source-specific use requires hazard deaggregation and path-conditioned terms inside the hazard integral.

The sampled domain covers events with $M_w \geq 5$ and fault distance no greater than 300 km. Incomplete plate-membership metadata restrict the anomalous-intensity term to a rule-based sensitivity analysis. Zhao receives shortest fault distance as rupture distance and AVS30 as VS30, and RotD50 serves as the closest available orientation-independent coordinate to its geometric-mean target. The two-event KiK-net waveform analysis verifies frequency-dependent surface-to-borehole response (Supplementary Fig. 7). Calibrating a nonlinear transfer model would require a larger event and input-motion range.

Across eight direct periods, the Japanese surface-record archive contains a repeatable station-associated field that changes matched long-period response spectra in both directions. Its average component is partly predictable from public site and location variables. Directional and short-distance tests identify the missing path dependence and define the additional information required before source-specific hazard application.

4 Methods

4.1 Strong motion data

We used the public `sub1-v2024` subset of the J-SHIS/NIED Strong Ground Motion Flat File[8, 9]. It contains 333,808 records with moment magnitude $M_w \geq 5$ and shortest fault distance no greater than 300 km. Record, source and site tables were joined through `smrec_id`, `eq_source_id` and `site_id`. K-NET and KiK-net provide the observations and metadata[10–12]. The official installation code identifies ground-surface and borehole sensors separately. The primary residual and spectrum analyses use ground-surface installations only; KiK-net borehole records enter only the waveform comparison.

MF2013 predicts the maximum norm of the two horizontal linear-oscillator responses, which is equivalent to RotD100. The primary analysis reads 5%-damped RotD100 spectral acceleration at 0.1, 0.2, 0.3, 0.5, 1.0, 2.0, 3.0 and 5.0 s[19]. Every value comes from its corresponding flatfile column. Records require positive spectral acceleration and distance, finite magnitude and model inputs, an MF2013 source class, positive AVS30 and a ground-surface installation code. These criteria retain 222,664 records at each period. RotD50 residuals are retained for a component sensitivity. At 3.0 s, the RotD50 and RotD100 station fields correlate at 0.9986 and differ by 0.018 $\log_{10}$ units at the 95th percentile (Supplementary Table 12).

4.2 MF2013 residuals

The MF2013 magnitude-distance prediction follows the published coefficients[13, 14]. The site calculation adds

$$G_d(T) = p_d(T) \log_{10} \left[\frac{\max(D_{l\,\min}(T), D_{1400})}{300} \right] \tag{1}$$

and

$$G_s(T) = p_s(T) \log_{10} \left[\frac{\min(V_{S\,\max}(T), AVS30)}{350} \right]. \tag{2}$$

Residuals are $r_{ij}(T) = \log_{10} Y_{ij}(T) - \log_{10} \widehat{Y}_{ij}(T)$ for station i and event j. The anomalous-intensity sensitivity follows the published northeastern and southwestern geographic and depth rules[31, 32]. Incomplete plate-membership flags restrict this calculation to a sensitivity analysis. The Philippine Sea Plate correction is an event-level constant at a given period[33]; the fitted event term absorbs this constant during station-effect estimation.

4.3 Event and station decomposition

For each period and MF2013 variant, we fit

$$r_{ij}(T) = \mu(T) + \delta E_j(T) + \delta S_i(T) + \epsilon_{ij}(T), \tag{3}$$

where μ is the global intercept, δE_j is the event term, δS_i is the station term and ϵ_{ij} is the record remainder. Alternating grouped least squares estimates the event and station terms. Record-count-weighted means of δE and δS are constrained to zero. We fit three variants at each period: the magnitude-distance backbone, the D1400/AVS30 site model and the site model with the anomalous-intensity sensitivity. Iteration stops when the largest parameter change is below 10^{-10} $\log_{10}$ units or after 5,000 iterations. All 24 period-model combinations converged; the largest final change was 9.99×10^{-11} and the largest iteration count was 1,161. Stations require at least 20 records for association and prediction analyses.

For the event holdout, earthquakes are randomly assigned to five groups with a fixed seed; the same partition is used at every period and for both ground-motion backbones. Each fold fits the two-way event-station decomposition separately to the four training groups and the held-out group. The paired train and test station vectors are each centred using their own record counts. Stations require at least 15 training records and five test records. We report the train-test station correlation and the test-record-weighted root mean squared error reduction relative to zero. Every training and test decomposition converges below 10^{-10} $\log_{10}$ units.

4.4 Alternative ground-motion-model sensitivity

The Zhao et al. 2006 active-crustal, interface and intraslab equations are evaluated with OpenQuake Engine 3.25.1[20, 21]. Source type selects the equation branch, the documented shortest fault distance supplies rupture distance, AVS30 supplies VS30, and the flatfile hypocentral depth and crustal rake complete the required context. OpenQuake defines Zhao for the geometric mean of two horizontal components; RotD50 is the closest orientation-independent response coordinate available in the flatfile and is used for this calculation. Predictions are evaluated at the same eight periods and surface-record sample as MF2013. The event-station decomposition, event holdout and spatial-block prediction retain the same thresholds and folds. This calculation measures backbone sensitivity within one observational archive.

4.5 Spatial station prediction

Two histogram gradient-boosting regressors predict the event-adjusted station term. The site-only model uses log-transformed VS10, VS20, VS30, AVS30, D1100, D1400, D1700, D2100 and basement depth, together with elevation, sensor depth and network. The site-and-location model adds longitude, latitude and the two volcanic-front distance fields. Missing numeric values are median-imputed within each training fold; network is mode-imputed and one-hot encoded.

Both models use squared-error loss, learning rate 0.04, 400 iterations, 15 maximum leaf nodes, 35 minimum samples per leaf and L2 regularisation 0.1. Five spatial blocks are formed by k-means clustering of standardised station coordinates. Each station is predicted only by the model trained on the other four blocks. Out-of-fold predictions are centred to zero using station record counts before conversion to multipliers $f_i(T) = 10^{\widehat{\delta S}_i(T)}$. The zero station term is the baseline. Reported errors are weighted by the number of records at each station.

4.6 Transfer and event-influence tests

Cross-network tests train on one surface network and evaluate the other. A common-feature variant excludes VS20, VS30 and network identity because their completeness differs between K-NET and KiK-net; it retains VS10, AVS30, sedimentary depths, elevation, sensor depth, coordinates and volcanic-front distances. Five fixed macroregions provide a second geographic leaveout test. Event influence is evaluated by removing each of the ten events with the largest 3.0 s record count and by removing all ten together at every period. Perturbed station terms are aligned to the full record-weighted zero reference before comparison.

4.7 Path and source stratification

The MF2013 RotD100 records are divided into four hypocentral-bearing sectors (000–090°, 090–180°, 180–270° and 270–360°), four shortest-fault-distance bins (0–50, 50–100, 100–200 and 200–300 km), and crustal, interplate and intraplate source classes. Bearing is the initial great-circle direction from the source horizontal coordinates to the station. Station-event graphs can become disconnected after stratification. Each connected component is solved by sparse least squares, and components with at least ten stations having ten records are aligned separately to the full station field. Five repeated event splits estimate within-stratum sampling reliability at 1.0, 2.0 and 3.0 s; each half requires five records per station.

For a persistent-field sensitivity, component-aligned station terms are averaged with equal weight across the available strata of each dimension. A station contributes when at least two azimuth sectors, distance bins or source classes are available. The physical and site-and-location models are refitted to each balanced field with the same five-block clustering procedure and hyperparameters used for the primary target. Strata receive equal weight in the target; prediction errors retain station record-count weights.

4.8 Station-model prediction intervals

For each held spatial block, signed out-of-fold errors from the other four blocks define record-weighted 5th and 95th percentiles. Adding these error quantiles to the held-block point prediction gives an empirical 90% station-term interval. Coverage is evaluated separately for each held block and then pooled by period. The lower and upper terms are converted to multipliers and propagated through the same response-spectrum calculation as the point estimate. These bounds cover station-model prediction error only.

4.9 Response spectrum propagation

We read official J-SHIS 2020 response-spectrum map ordinates at the same eight periods and at 50-year exceedance probabilities of 2%, 5%, 10% and 39%[27, 29, 30]. The maps define spectral acceleration on engineering bedrock with $V_S = 400$ m s^{-1}. For station i, the surface-reference factor is

$$F_{\mathrm{surf},i}(T) = 10^{G_s(AVS30_i,T)-G_s(400,T)}. \tag{4}$$

The ergodic surface ordinate is $S_{\mathrm{surf},i} = S_{400,i}F_{\mathrm{surf},i}$, and the station-adjusted ordinate is

$$S_{\mathrm{adj},i}(T) = S_{\mathrm{surf},i}(T)10^{\widehat{\delta S}_i(T)}. \tag{5}$$

Only out-of-fold station predictions enter this calculation. Quantiles are evaluated over the 1,628 matched surface stations. The correction represents the average source-path distribution in the strong-motion archive. City cases use the nearest matched station by great-circle distance and are reported as station examples.

4.10 KiK-net waveform check

A separate waveform check uses 460 paired KiK-net surface and downhole records from two earthquakes. Traces are demeaned and detrended, horizontal amplitudes are calculated from the two horizontal components, and surface-to-downhole Fourier amplitude ratios are summarised from 0.2 to 20 Hz. This analysis verifies frequency-dependent surface response in the network records and is excluded from the residual model and response-spectrum propagation.

4.11 Artificial intelligence use

The manuscript and code were written by the authors. ChatGPT was used only for language and formatting revision and code verification. The authors executed the code, inspected the source data and outputs, verified the reported values and references, and take responsibility for the study and final manuscript. No generative-AI image is included.

4.12 Data availability

The strong-motion flatfile is available under DOI `https://doi.org/10.17598/NIED.0032`. MF2013 coefficients and documentation are available at `https://www.j-shis.bosai.go.jp/labs/mf2013/en/`. J-SHIS response-spectrum map products are available from `https://www.j-shis.bosai.go.jp/`. Raw third-party records and map archives are not redistributed. Derived tables used in the main figures and Supplementary Information are included in the project repository at `https://github.com/zhouhaoyiu/japan-strong-motion-site-terms`.

4.13 Code availability

Analysis and figure-generation code is available at `https://github.com/zhouhaoyiu/japan-strong-motion-site-terms`. The repository includes environment specifications, reproduction commands and workflows for the primary and Zhao residual calculations, uncertainty propagation, transfer tests, path stratification and balanced-field sensitivity.

Author contributions

H.Z. designed the analysis, processed the public datasets, performed the computations, prepared the figures, and wrote the manuscript. Q.M. supervised the study and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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